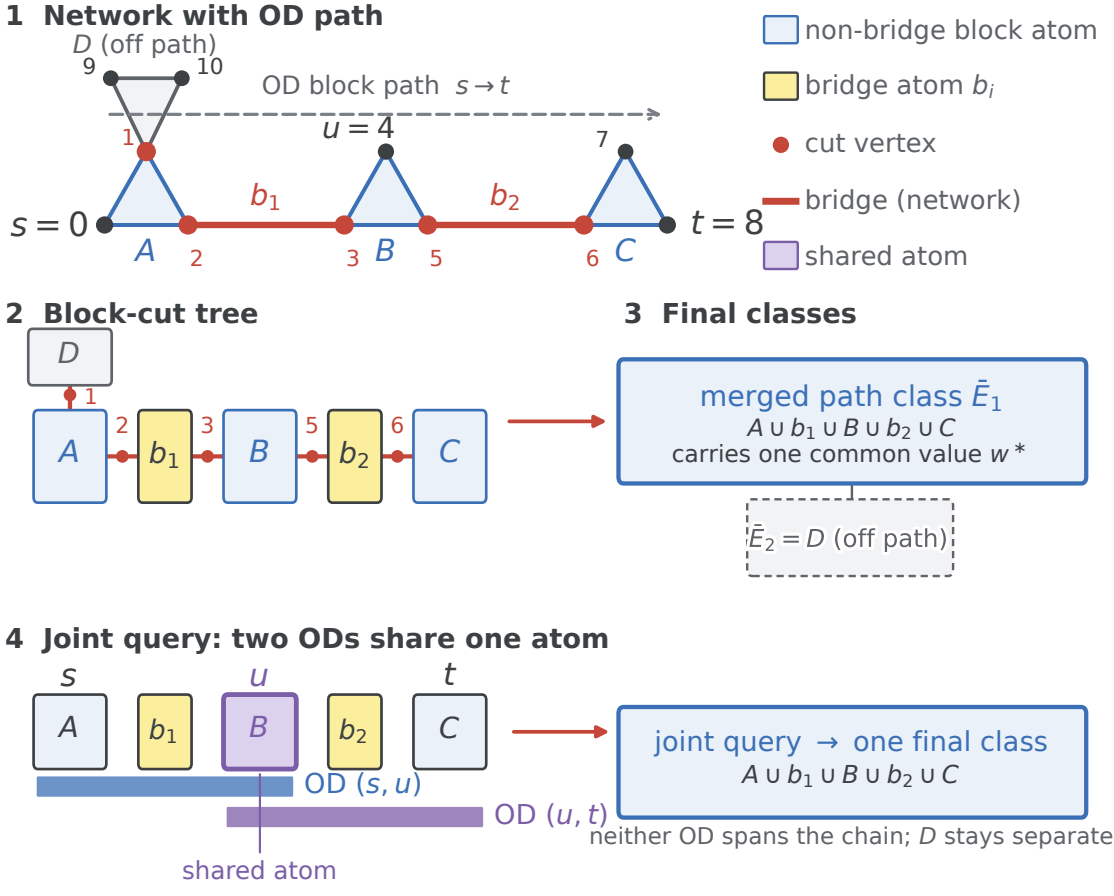
1 Network with OD path
D (off path)
9
10
OD block path s → t
1
u = 4
7
s = 0
b1
b2
t = 8
A
2
3
B
5
6
C
non-bridge block atom
bridge atom bi
cut vertex
bridge (network)
shared atom
2 Block-cut tree
D
1
A
2
b1
3
B
5
b2
6
C
3 Final classes
merged path class Ē1
A ∪ b1 ∪ B ∪ b2 ∪ C
carries one common value w*
Ē2 = D (off path)
4 Joint query: two ODs share one atom
s
u
t
A
b1
B
b2
C
OD (s, u)
OD (u, t)
shared atom
joint query → one final class
A ∪ b1 ∪ B ∪ b2 ∪ C
neither OD spans the chain; D stays separate